

MILLVILLE, NJ, USA

WMO: 724075

Lat: 39.366N

Lon: 75.078W

Elev: 60

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13735

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.7	15.4	-3.4	4.6	15.4	1.2	5.8	19.5	25.1	34.3	22.7	32.6	5.7	300	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	91.9	75.2	89.3	74.2	86.9	73.1	78.4	86.8	77.1	84.9	75.9	83.0	8.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.1	136.4	82.1	74.8	130.9	80.8	73.6	125.4	79.6	41.8	86.9	40.4	85.0	39.3	83.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.4	16.4	DB	1.7	96.2	5.1	3.0	-1.9	98.4	-4.9	100.2	-7.8	101.9	-11.5	104.1	(4)
(5)			WB	1.1	80.8	4.9	1.4	-2.4	81.8	-5.3	82.6	-8.1	83.4	-11.6	84.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	33.1	35.0	42.2	52.7	62.0	71.3	76.3	74.6	68.0	56.5	45.8	37.8	(6)
(7)		DBStd	17.13	10.53	9.53	9.20	8.55	7.61	6.48	5.11	5.01	6.52	8.18	8.55	9.17	(7)
(8)		HDD50	1902	533	427	275	64	3	0	0	0	0	28	180	392	(8)
(9)		HDD65	4835	990	839	706	381	153	20	0	1	42	286	575	842	(9)
(10)		CDD50	3629	9	9	35	145	374	639	815	763	540	229	56	15	(10)
(11)		CDD65	1087	0	0	1	13	59	209	350	299	133	22	1	0	(11)
(12)		CDH74	9616	0	0	14	144	557	1855	3343	2593	972	135	3	0	(12)
(13)		CDH80	3267	0	0	2	36	153	627	1313	873	240	23	0	0	(13)
(14)	Wind	WSAvg	6.6	7.6	7.7	8.2	8.0	6.6	5.9	5.2	4.8	5.6	6.0	6.5	6.8	(14)
(15)	Precipitation	PrecAvg	41.6	3.2	3.1	4.1	3.6	3.6	3.1	3.5	4.2	3.2	3.0	3.2	3.6	(15)
(16)		PrecMax	58.6	7.9	7.6	7.0	7.6	8.2	7.2	12.9	11.4	7.7	6.5	8.6	9.3	(16)
(17)		PrecMin	24.0	0.2	0.6	1.3	0.6	0.8	0.4	0.6	0.9	0.6	0.1	0.5	0.4	(17)
(18)		PrecStd	7.7	1.8	1.7	1.7	1.7	2.0	2.0	2.4	2.8	1.7	1.7	2.2	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	67.7	74.6	84.7	89.1	93.6	96.2	94.4	90.3	82.9	72.4	67.0	(19)
(20)			MCWB	58.0	57.4	59.6	65.2	71.4	75.9	76.9	77.1	73.9	70.5	62.8	60.6	(20)
(21)		2%	DB	59.2	60.1	67.9	78.3	84.4	89.8	92.5	90.3	85.8	77.8	67.3	61.0	(21)
(22)			MCWB	55.5	52.5	56.1	62.2	69.5	73.7	75.6	75.3	72.5	68.4	59.4	56.8	(22)
(23)		5%	DB	54.6	55.3	62.2	72.7	80.5	86.6	89.8	87.6	82.4	73.8	64.0	57.1	(23)
(24)			MCWB	50.9	48.8	52.6	59.8	67.5	72.5	75.1	73.9	70.8	65.7	58.4	53.9	(24)
(25)		10%	DB	49.5	50.4	57.4	67.8	76.3	83.5	87.1	84.8	79.6	70.5	60.8	53.1	(25)
(26)			MCWB	45.3	44.7	49.8	57.0	65.2	71.2	73.8	72.6	69.2	63.5	55.8	49.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.0	59.9	62.7	68.4	74.0	78.0	80.3	80.4	77.4	73.7	65.7	62.7	(27)
(28)			MCD B	62.3	64.6	72.5	78.6	85.1	88.4	89.8	88.8	84.3	77.6	69.2	65.6	(28)
(29)		2%	WB	56.1	54.7	58.4	65.0	71.6	76.3	78.5	78.3	75.5	71.0	62.6	58.3	(29)
(30)			MCD B	58.9	58.0	64.3	74.4	80.9	85.6	87.4	86.1	80.7	76.2	65.2	60.5	(30)
(31)		5%	WB	51.2	50.0	55.0	62.4	69.6	75.0	77.4	76.8	74.1	68.1	59.7	54.5	(31)
(32)			MCD B	53.9	54.0	59.8	70.0	77.5	83.2	85.6	83.5	79.1	72.0	63.0	56.7	(32)
(33)		10%	WB	46.0	45.2	51.4	59.4	67.6	73.4	76.2	75.4	72.4	65.0	56.5	50.1	(33)
(34)			MCD B	48.6	50.1	56.8	66.0	74.3	80.4	83.7	81.4	77.2	69.0	59.9	52.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.2	18.7	20.1	22.0	20.9	19.9	18.8	18.7	19.7	20.6	20.0	17.7	(35)
(36)			MCDBR	21.9	24.6	27.0	27.6	24.8	22.4	21.1	20.7	21.5	22.9	22.2	20.7	(36)
(37)			MCWBR	18.3	18.3	17.1	14.5	12.2	9.7	7.9	8.3	10.2	13.4	15.3	17.8	(37)
(38)		5% WB	MCD BR	20.7	22.5	23.1	22.5	21.1	19.6	18.1	17.2	16.9	18.1	17.7	18.8	(38)
(39)			MCWBR	18.8	19.1	17.5	13.7	11.7	9.4	7.9	7.8	9.2	12.3	14.5	17.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.318	0.367	0.392	0.438	0.497	0.508	0.501	0.417	0.361	0.329	0.312	(40)	
(41)		taud	2.472	2.458	2.362	2.322	2.223	2.104	2.087	2.093	2.302	2.463	2.510	2.523	(41)	
(42)		Ebn at Noon	272	287	282	280	268	252	248	246	261	268	265	265	(42)	
(43)		Edh at Noon	25	29	36	39	44	50	51	49	37	29	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	638	922	1225	1577	1763	1931	1942	1711	1405	1021	731	562	(44)	
(45)		RadStd	55	101	99	123	211	150	90	110	140	102	65	55	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+234	+137

Nomenclature: See separate page